LiDAR 3D Object Annotation and Quality Assurance Tracker

Project Overview

The LiDAR 3D Object Annotation and Quality Assurance Tracker is an Excel-based portfolio project designed to organize LiDAR annotation metadata and track quality-review activities for autonomous driving and robotics applications.

LiDAR stands for Light Detection and Ranging. A LiDAR sensor uses laser measurements to create a three-dimensional representation of the surrounding environment, commonly called a point cloud.

In autonomous driving applications, objects such as vehicles, pedestrians, cyclists, and traffic cones may be labelled using 3D bounding boxes. These boxes can contain information about an object's position, dimensions, and orientation.

This project uses a structured Excel workbook to track representative annotation records, document quality issues, and summarize annotation progress through a dashboard.

Important: This project is an Excel-based annotation tracking and QA reporting tool. It does not directly annotate raw LiDAR point clouds or generate 3D bounding boxes. The workbook uses fictional sample records for demonstration.

Objectives

- Organize LiDAR 3D object annotation metadata.
- Track object classes, coordinates, bounding-box dimensions, and orientation.
- Monitor annotation completion and review status.
- Document quality issues and corrective actions.
- Summarize annotation and QA results.
- Create a dashboard for progress monitoring and reporting.

Tools and Technologies

- Microsoft Excel
- Excel formulas and functions
- Data validation
- Sorting and filtering
- Charts and KPI dashboards
- Structured data management

Key Features

1. LiDAR Annotation Metadata

Organizes representative annotation information, including:

- Object ID and scene ID
- Object class
- X, Y, and Z center coordinates
- Bounding-box length, width, and height
- Yaw angle
- Annotator and annotation status

Coordinate units and conventions depend on the source dataset and annotation tool.

2. Quality Assurance Tracking

Provides fields for recording quality-review results and documenting issues such as:

- Incorrect object classification
- Poor bounding-box fit
- Dimension mismatches
- Orientation inconsistencies
- Missing or incomplete annotation records
- Objects requiring further review

3. Annotation Progress Monitoring

Tracks the status of annotation records and helps identify tasks that are completed, pending, or awaiting review.

4. KPI Dashboard

Summarizes annotation progress, object-class distribution, and quality-review outcomes using Excel-based reporting.

5. Summary Reporting

Organizes annotation and QA information to support progress tracking and data-quality monitoring.

Project Workflow

1. Review the sample LiDAR annotation records.
2. Examine object classes and 3D metadata fields.
3. Check that required fields are populated.
4. Review annotation and QA status.
5. Document quality issues and recommended corrective actions.
6. Examine the dashboard and summary sheets.

Dataset Information

The workbook contains fictional sample annotation records created for learning and portfolio demonstration. It is not a real LiDAR point-cloud dataset and does not represent actual company or production results.

Expected Learning Outcomes

- Understanding LiDAR annotation metadata
- Learning common 3D annotation quality issues
- Practising structured data validation
- Tracking QA results and annotation progress
- Building Excel-based KPI dashboards
- Understanding quality monitoring in autonomous driving data workflows

Project Files

- LiDAR_3D_Object_Annotation_QA_Tracker.xlsx — Excel workbook containing the sample annotation records, QA tracking, summary information, dashboard, and project guide.

How to Use

1. Download the Excel workbook from this repository.
2. Open it in Microsoft Excel.
3. Read the Project Guide sheet.
4. Explore the LiDAR Annotations and QA Review sheets.
5. Review the Summary and Dashboard sheets.

6. Examine how the workbook organizes annotation metadata and quality-review information.

Limitations

This workbook tracks annotation metadata and QA results. It does not visualize point clouds, create 3D cuboids, or automatically validate the geometric accuracy of bounding boxes.
Those tasks require raw LiDAR data and specialized annotation or visualization software.

Conclusion

This project demonstrates how structured metadata, quality-review records, and Excel dashboards can support the monitoring of LiDAR annotation workflows. It is relevant to robotics data annotation, autonomous driving data operations, and quality assurance roles.